Wave and Optics
Part-2

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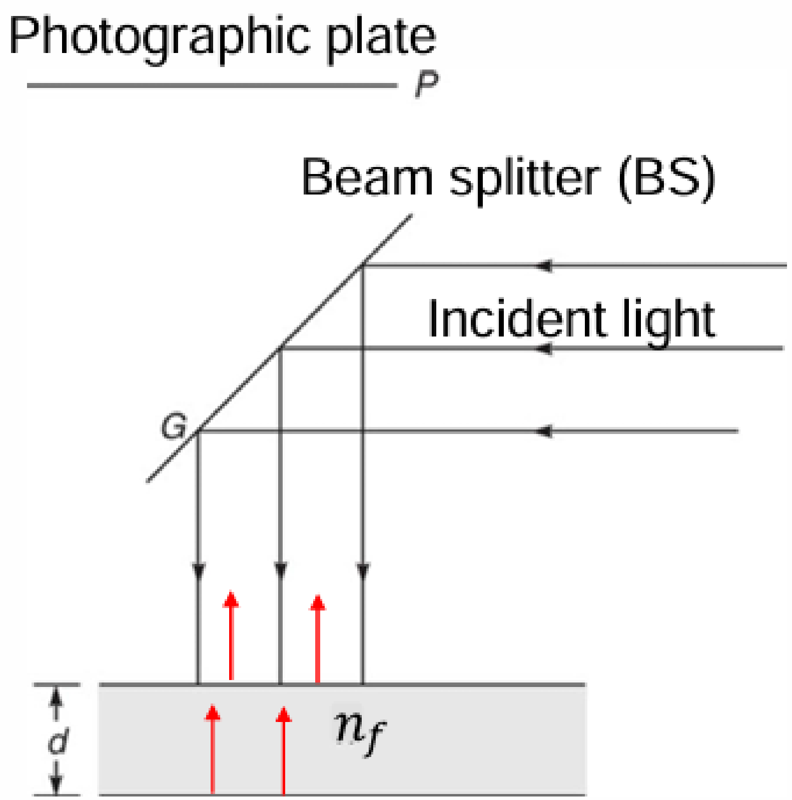
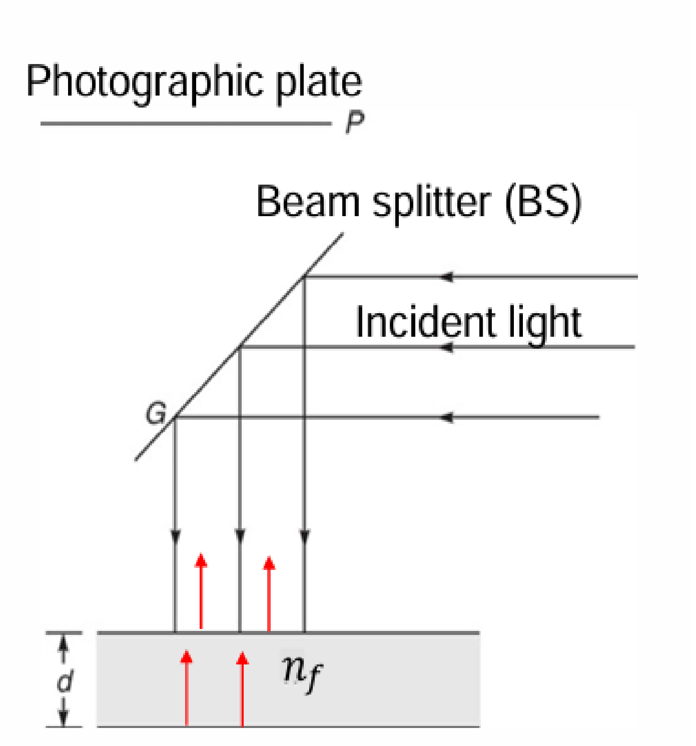
Interference pattern produced by division of amplitude

In this case, a plane wave is allowed to fall on thin film and then wave reflected from the upper surface interferes with the wave reflected from the lower surface to produce interference pattern.



- Wave reflected from the upper and lower surface of the film will interfere at the photographic plate P
- Optical path difference from the one reflecting from the upper surface and the one from the lower surface ?
- $2n_f d$ because the film thickness d is traversed twice
- Additionally the beam reflected from the upper surface undergoes an additional phase change of π as it is reflected from the interface of air and film of higher refractive index.



The effective wavelength in a medium of refractive index n_f is λ/n_f

Clearly wave reflected from lower surface of film traverses an additional path of $2n_f d$

If the film is placed in air, in such a case wave reflected from upper surface of film will undergo a sudden change in phase of π

Since for constructive interference, phase difference between the two interfering beams should be $2m\pi$

Net $\Delta\varphi$?

$$\Delta\varphi = k_o n_f \times (path\ length\ difference)$$

$$\Rightarrow \frac{2\pi}{\lambda_o} \times n_f \times 2d - \pi = 2m\pi \text{ (For Maxima)}$$

Additional π phase shift accumulated by the upper beam due to reflection from air-glass interface

$$\frac{2\pi}{\lambda_o} \times n_f \times 2d = 2\pi\left(m + \frac{1}{2}\right) ; m = 0,1,2,\ldots$$

$\Rightarrow$ for constructive interference

$$2n_f \times d = \left(m + \frac{1}{2}\right)\lambda_o ; m = 0,1,2,\ldots$$

This is because the wave is reflected out of phase when it tries to enter a medium with a slower speed of light.

For destructive interference :

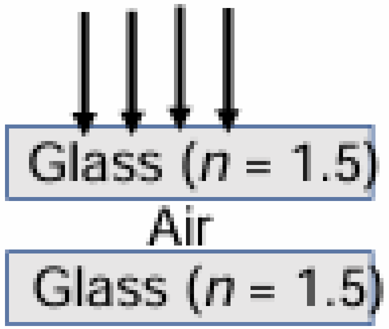
$$\frac{2\pi}{\lambda_o} \times n_f \times 2d - \pi = (2p+1)\pi\ ; \quad p = 0,1,2,\ldots$$

$$\Rightarrow 2\pi \times n_f \times 2d = (2p\pi + \pi + \pi)\lambda_o = 2\pi \times (p+1)\lambda_o\ ; p = 0,1,2,\ldots$$

$$\Rightarrow 2n_f \times d = (p+1)\lambda_o\ ; p = 0,1,2,\ldots \equiv m\lambda_o\ ; m = 1,2,\ldots$$

$\Rightarrow 2n_f d = m\lambda_o$; m $= 1,2,\ldots$: for destructive interference

If an air film is sandwiched between two glass plates :

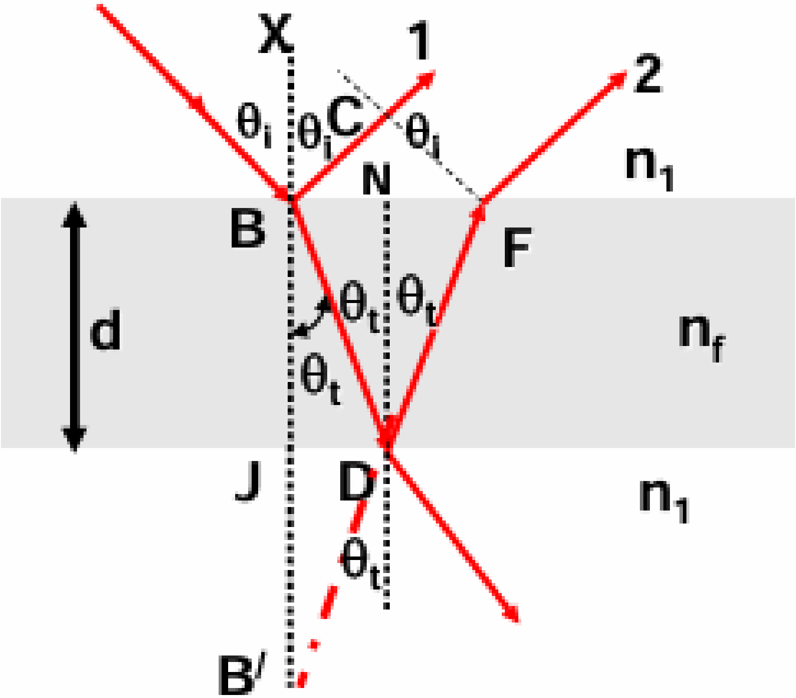


Extra phase change of π is only at the air-glass interfaces

$\Rightarrow$ Same interference condition

Oblique incidence (cosine law) :

Geometrical path difference Δ between the wave reflected from the upper surface (ray 1) of the film(BJ) and the one (ray 2) reflected from the lower surface (BD + DF) which interfere can be shown to be given by ,



$$\Delta = 2d\cos\theta_t$$ (see θ_t on the fig)

⇒ Optical phase difference φ :

$$\frac{2\pi}{\lambda_o} n_f \Delta - \pi = \frac{2\pi}{\lambda_o} \times 2dn_f \cos\theta_t - \pi$$

…Due to π phase shift acquired by the ray (1)

Thus, in case of oblique incidence for constructive interference,

Optical phase difference should be :

$$\varphi = \frac{2\pi}{\lambda_o} \times 2dn_f \cos\theta_t - \pi = 2m\pi \,; m = 0,1,2, \ldots$$

$$\Rightarrow \frac{2\pi}{\lambda_o} \times 2dn_f \cos\theta_t = 2\pi\left(m + \frac{1}{2}\right)$$

$$\left.\begin{aligned} \Rightarrow 2dn_f \cos\theta_t (\equiv \Delta) &= \left(m + \frac{1}{2}\right)\lambda_o \ : \text{maxima} \\ &= m\lambda_o \ : \text{minima} \end{aligned}\right\} \text{cosine law}$$

Application of thin film interference

For near-normal incidence ($\Rightarrow \cos\theta_t \cong 1$) of light at a glass-air interface (air R.I =1), reflectivity of glass (n_g) can be shown to be given by

$$R = \left(\frac{n_g - 1}{n_g + 1}\right)^2$$

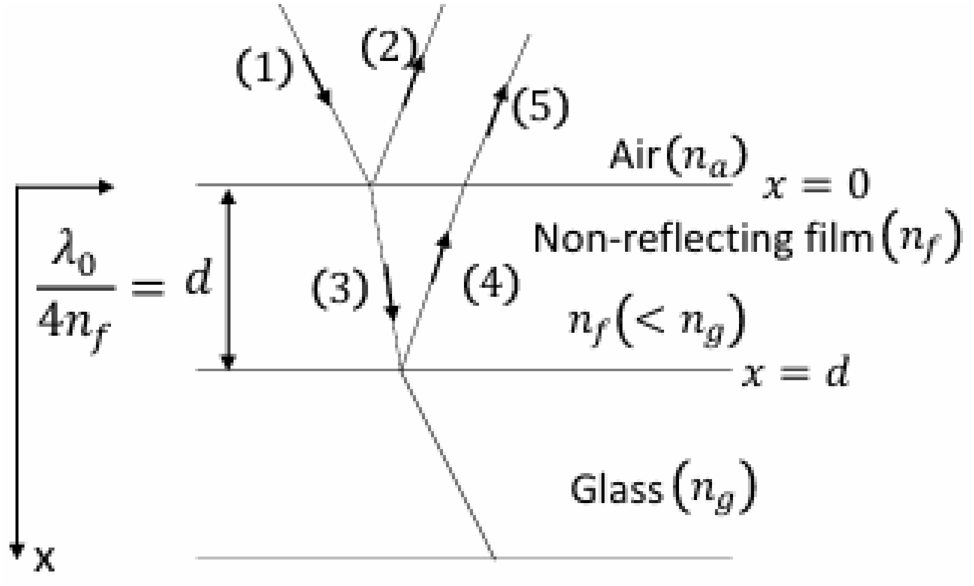
For crown glass e.g. window glass, $n_g = 1.5$

$$\Rightarrow R = \left(\frac{1.5-1}{1.5+1}\right)^2 = \left(\frac{0.5}{2.5}\right)^2 = \left(\frac{1}{5}\right)^2 = (0.2)^2 = 0.04$$

4% of incoming light gets reflected and hence lost in transmission !

$\Rightarrow$ a glass window though meant for 100% transmission of light transmits only 96% of incident light

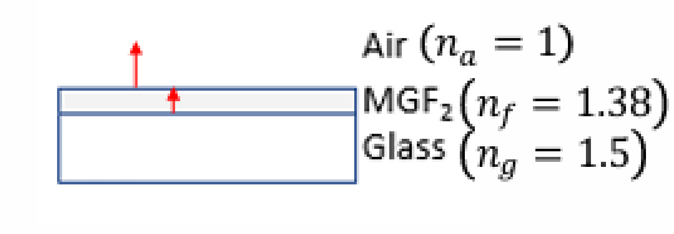
Consider non-reflecting films :



If we wish to coat a non-reflecting film on crown glass

⇒ Reflected rays (2) and (5) should interfere destructively

Consider coating of a MgF_2 film of R.I. 1.38 on glass



⇒ There will be a π phase shift acquired by reflected light at both air- MgF_2 film (R.I. 1.38) interface and at the MgF_2 film – glass (R.I. 1.5) interface

⇒ Conditions for maxima and minima will be reversed vis-à-vis air-film-air case

Thus conditions for destructive interference between reflected rays (2) and (5) in case of air- MgF_2 film – Glass, we get from

$$2n_f d = \left(m + \frac{1}{2}\right)\lambda_0 \;;\; m = 0,1,2,\ldots$$

For m = 0 :

$$2n_f d = \frac{\lambda_0}{2} \Rightarrow d = \frac{\lambda_0}{4n_f}$$

Thus, for $\lambda \sim 5 \times 10^{-5}\ cm$,

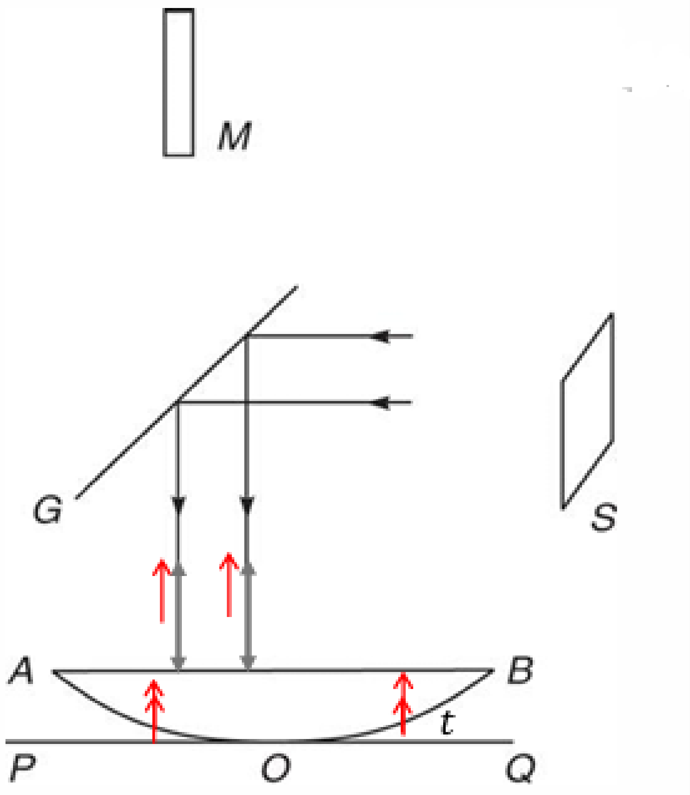
$$d \cong \frac{5 \times 10^{-5}}{4 \times 1.38}$$

$$\approx 9 \times 10^{-6}\ cm = 0.09 \times 10^{-4}\ cm$$

$$= 0.09\ \mu m$$

Interference by Wedge films (i.e., a film with two non-parallel reflecting surfaces): Will be part of your home assignment. You will workout yourself.

Formation of colours in the films: Will be part of your home assignment. You will workout yourself.



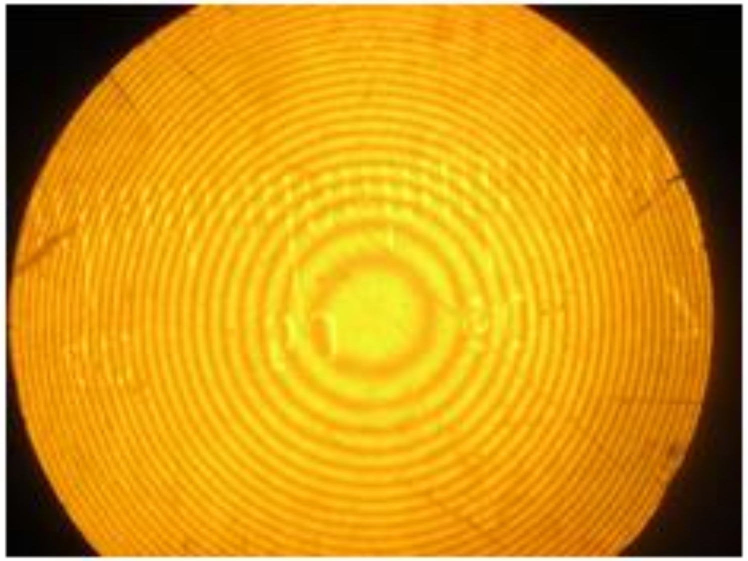
Interference takes between light reflected from AOB and POQ

$$\therefore \text{ for maxima : } 2t = \left(m + \frac{1}{2}\right)\lambda_o\,; \quad m = 1,2,\ldots$$

$$\text{for minima : } 2t = m\,\lambda_o \quad ; \; m = 1,2,\ldots$$

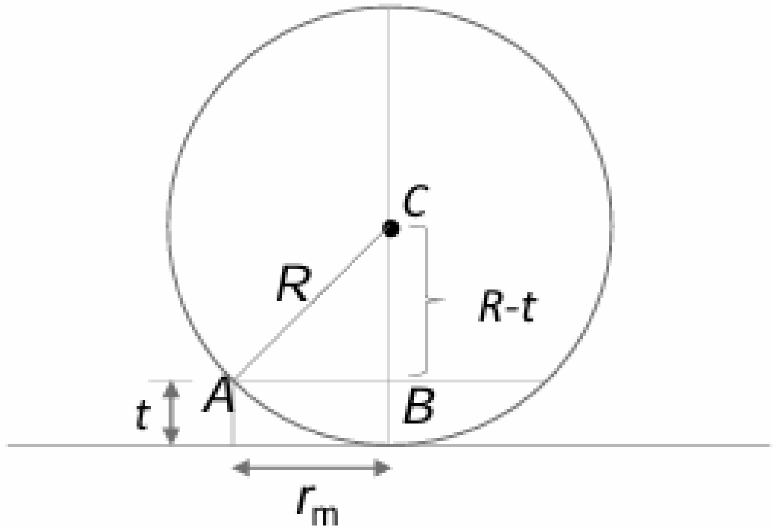
Due to the spherical surface of the lens, t will be constant over a circle with o as its centre $\Rightarrow$ we will get concentric dark and bright fringes in the form of rings

The fact that the wave is reflected from air to glass surface introduces a phase shift of π



Concentric dark and bright rings
Known as Newton's ring

Radius of the m^{th} dark ring :



From the figure in the triangle ABC

$$(R-t)^2 + {r_m}^2 = R^2$$

$$\Rightarrow R^2 = {r_m}^2 + R^2 - 2tR + t^2$$

$$\Rightarrow {r_m}^2 \cong t(2R - t)$$

Typically, $R \sim 100cm\,; t \sim 10^{-3}\ cm$

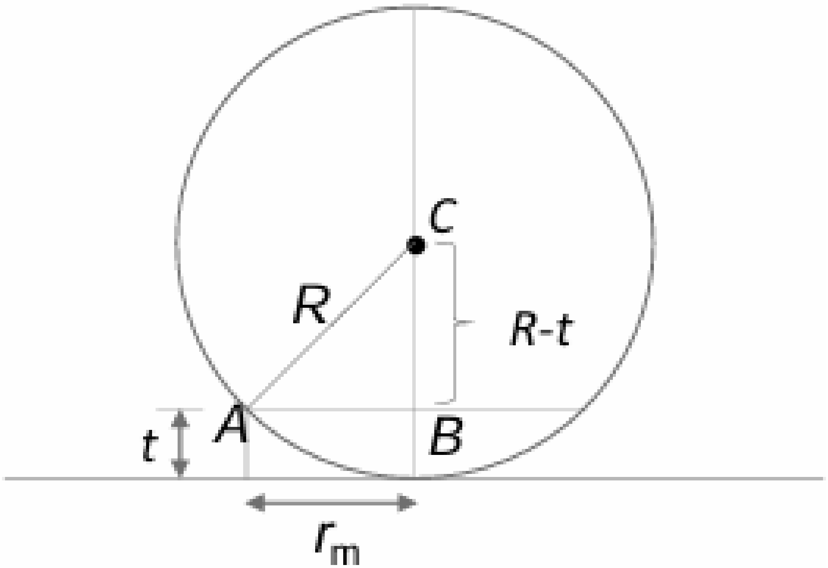
$\Rightarrow$ t^2 can be neglected relative to $2R$

$$\Rightarrow \qquad 2t = \frac{{r_m}^2}{R} = m\lambda_o \qquad\qquad \Rightarrow r_m = \sqrt{mR\lambda_o}\ ; m = 1,2,\ldots$$

(for m^{th} dark ring)

$\Rightarrow$ Radii of the dark ring vary as square root of natural numbers ;

its diameter (D_m) will be $2\sqrt{mR\lambda_o}$



In the expt., diameters of m^{th} and $(m+p)^{th}$ rings are measured (p~10) and from the following relation source wavelength λ_o is measured.

$$(D_{m+p})^2 - (D_m)^2 = 4(m+p-m)R\,\lambda_o = 4\mathrm{pR}\lambda_o$$

$$\Rightarrow \boxed{\frac{(D_{m+p})^2 - (D_m)^2}{4pR} = \lambda_o}$$

Note:

1. In Newton's ring experiment, if the light consist of two closely spaced wavelengths λ_1 and λ_2
(like sodium lamp, λ_1 = 589.0 nm and λ_2=589.6 nm), then if the lens is separated by a distance t_0

$t_0 = \lambda_1 \lambda_2 / 4(\lambda_2 - \lambda_1)$ the interference fringes will be washed out.

The fringes will reappear when the distance is $2t_{0.}$

2. Fringe width changes with distance from the center of lens

Will be part of your home assignment and self leairning

Thanks